

Microbial biomass (Novacq™) stimulates feeding and improves the growth performance on extruded low to zero-fishmeal diets in tilapia (GIFT strain)



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ARTICLE INFO

Keywords:

Microbial biomass
Palatability
Sustainability
Fishmeal replacement
Tilapia nutrition

ABSTRACT

Worldwide tilapia production is rapidly growing and becoming increasingly reliant on the provision of cost-effective sustainable formulated feeds. We assessed the capacity for various inclusion rates of the microbial biomass-based ingredient Novacq™ (0%, 2.5%, 5%, 10%) at the expense of wheat flour to improve growth performance in diets at 10% fishmeal (FM), as well as the growth response at 5% and 0% FM, with or without the addition of 10% Novacq™ in iso-nitrogenous (CP = 34.5–36.1%) extruded diets for GIFT tilapia fingerlings. A total of 540 fish (mean weight $\pm$ SD: 11.4 $\pm$ 1.8 g) were reared for 6 weeks on 8 experimental diets and a commercial benchmark diet (6 replicate tanks on each diet) in a recirculated freshwater system. All extruded diets containing Novacq™ outperformed the benchmark commercial diet. Total replacement of fishmeal (0% FM) with soybean meal resulted in a significant decrease of around 14.5% in daily weight gain ($P < .05$), but 50% replacement (5% FM) did not significantly affect weight gain ($P > .05$). Weight gain increased in parallel with increasing inclusion rates of Novacq™, with weight gains of 7.8%, 23.6% and 34.5% in relation to an isoenergetic and isonitrogenous control diet, at 2.5%, 5% and 10% Novacq™ inclusion, respectively. Novacq™ at 10% in diets containing 5% and 0% fishmeal also significantly increased the growth performance of tilapia, producing a net improvement in weight gain of 19.5% and 15.5% against the 10% fishmeal control diet, respectively. Feed intake mirrored the growth trends. A 10% inclusion of Novacq™ resulted in approximately 33% increase in feed intake at all levels of fishmeal inclusion. Daily feed intake measurements, similar FCRs (1.05–1.18) and nutrient retention efficiencies (Protein RE = 37–46%; Energy RE = 23–30%), indicated Novacq™ principally stimulated feeding with no adverse effect on nutrient use. The use of Novacq™ as a feed additive can more than compensate for the negative impact of fully eliminating fishmeal and is likely to improve the sustainability value of tilapia diets.

1. Introduction

Cichlids, particularly tilapias, are the second most farmed fish worldwide, with global production of 5.67 million tonnes valued at 8.9 billion USD in 2015 (FAO, 2018). The GIFT strain (Genetically Improved Farmed Tilapia) is a selectively bred Nile tilapia for improved growth performance, *Oreochromis niloticus*, accounting for 80% of total tilapia seed production in China, 75% in Thailand and 40% in the Philippines in 2010 (Sukmanomon et al., 2012). Worldwide tilapia production is rapidly growing and becoming increasingly reliant on the provision of cost-effective sustainable formulated feeds. The nutrient and feeding requirements of some of the improved tilapia strains are not fully understood, and considering feed cost is the major expenditure in tilapia farming, more research effort in developing species (strain)-

specific diets is warranted (Ng and Romano, 2013).

The replacement of fishmeal and fish oil by more sustainable and cheaper alternatives in diets for tilapia has been one major research focus for the last decade. Currently, fishmeal and fish oil are included between 0% and 20% and 0% and 10%, respectively, in commercial diets. Fishmeal inclusion rate is predicted to decline to 1% by 2020 due to its rising cost and unpredictable reliability (Ng and Romano, 2013). There have been many studies to evaluate the replacement of fishmeal in practical diets for tilapia with less expensive, locally-available plant and animal meals (El-Saidy and Gaber, 2002; Herath et al., 2016; Koch et al., 2016; Ng and Romano, 2013; Shiao et al., 1989). Successful complete replacement of fishmeal with soybean meal (SBM), soy protein concentrate (SPC), and poultry by-product meal (PBM) with addition of essential amino acids (EAA such as Met, Lys) and/or

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<https://doi.org/10.1016/j.aquaculture.2018.11.052>

Received 8 October 2018; Received in revised form 27 November 2018; Accepted 27 November 2018

Available online 28 November 2018

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phosphorus has been achieved in several short-term studies on tilapia fingerlings (Ng and Romano, 2013). However, a recent study showed tilapia performance on a diet containing SBM + EAA as the sole source of protein to be poorer than on diets also containing 15–30% PBM and a control diet containing 20% fishmeal (Koch et al., 2016). By all accounts, fishmeal can be replaced with a careful selection of plant and/or animal by-product meals in tilapia diets but the resulting growth is similar or lower to when fishmeal is added, and other metrics such as FCR and protein retention may be affected negatively (Herath et al., 2016; Koch et al., 2016).

Microbial biomass produced using different biotechnological processes can be used to supplement aquafeeds in order to improve performance and sustainability. The use of whole microbial biomass has had in recent times varying degrees of success in prawns (Kuhn et al., 2010; Anand et al., 2014; Qiu and Davis, 2018; Qiu et al., 2018) and tilapia (Avnimelech, 2007; Azim and Little, 2008; Caldini et al., 2015; Grassi et al., 2018). These studies have shown that large variation in nutrient content as well as the form of the microbial biomass can have an effect. For example, tilapia fed dry microbial biomass grew slower than on wet biofloc biomass (Caldini et al., 2015). Studies by Glencross et al. (2014, 2015) indicated that a microbial biomass-based dry ingredient, marketed as Novacq™, can provide a sustainable solution to entirely eliminate the need for fishmeal in *P. monodon* feeds. Interestingly, unlike bacterial biomass rich in protein and useful as a direct protein replacer (Qiu and Davis, 2018; Qiu et al., 2018), Novacq™ tends to contain little protein and is likely to instead provide bioactive micronutrients which enable plant-based protein sources to perform similarly to fishmeal (Glencross et al., 2015). Inclusions of 5% and 10% Novacq™ have also shown to increase growth at a range of fishmeal and protein inclusion levels by as much as 60% (Glencross et al., 2015). This was confirmed in a subsequent prawn experiment (Arnold et al., 2016), however Novacq™ did not have an obvious effect on the measured reproductive performance of *P. monodon* (Goodall et al., 2016).

It is possible that among various physiological roles bioactive compounds found in microbial biomass act as feed attractants but feed consumption of messy-feeders like prawns is often difficult to measure accurately (Glencross et al., 2015; Arnold et al., 2016; Wade et al., 2018). Accurate feeding and assessment of feed intake by fish is one of the more difficult aspects of aquaculture nutrition research (Glencross et al., 2007). Characterising the palatability of ingredients is important, because regardless of their nutrient composition and digestibility, it can have a major impact on its usefulness in a feed formulation. The most common physicochemical properties of feeding attractants and stimulants for fish are non-volatile, low molecular weight, nitrogen-containing, amphoteric, water soluble, stable to heat treatment, and of broad biological distribution. These properties are consistent with those of free amino acids and related nitrogenous substances like nucleotides, nucleosides, and quaternary ammonium bases (de la Higuera, 2007).

The Novacq™ ingredient is now in full-scale commercial production for shrimp globally but despite the large demonstrated success in shrimp, the potential application in other species including finfish has not been tested. Considering the potential for microbial biomass to be successful as a feed additive in an omnivorous fish species like tilapia, we assessed the capacity for various inclusion rates of the feed additive Novacq™ to improve growth performance in extruded diets containing

10% fishmeal. We also assessed the growth response at 5% and 0% fishmeal, with or without the addition of 10% Novacq™ in extruded diets for GIFT tilapia fingerlings.

2. Material and methods

2.1. Culture system and fish population

The growth experiment was conducted in the laboratories of the WorldFish headquarters in Penang, Malaysia, using GIFT (Genetically Improved Farmed Tilapia) Nile tilapia, *Oreochromis niloticus*. A total of 54 × 10 L transparent aquaria (tanks) on a recirculated freshwater system were used. Water flow was regulated at 2 L min⁻¹ tank⁻¹, and recirculated through a series of filtration media including a screen filter (90 µm) to remove large solids and a fibreglass sand filter (Waterco sand filter) with mixed media (MM1 Mixed media Pentair). Around 30% of water in this system was renewed every 3–4 days depending on water quality. The water quality over the entire experiment was kept as follows: mean ± S.D. water temperature = 27.1 ± 0.8 °C; D.O. 7.0 ± 1.1 mg L⁻¹; pH = 6.8 ± 0.3; NH₃ < 0.5 ppm; NO₂ < 1.0 ppm; NO₃ < 80 ppm. A total of 1500 mixed sex fish fingerlings were obtained from the WorldFish GIFT breeding centre in Jitra, Malaysia. Fish were maintained and anaesthetised in accordance with the 'Australia Code for the care and Use of Animals for Scientific Purposes' 8th edition. The stock was graded by weight to only include fish within one standard deviation of the mean to use for the experiment. A total of 10 fish per tank (mean weight 11.4 ± 1.8 g ± SD) were randomly allocated. Three samples of 10 pooled initial fish were taken for subsequent whole carcass composition analyses.

2.2. Dietary treatments and composition

A total of eight isonitrogenous and isoenergetic experimental diets were formulated to address nutritional requirements of juvenile tilapia (Ng and Romano, 2013). The chemical composition of a commercial benchmark diet (Cargill 6113 floating starter 4 × 5 mm) and the ingredients used in this study can be found in Table 1. Dietary treatments consisted in four diets containing 10% fishmeal with incremental level of Novacq™ (0%, 2.5%, 5% and 10%), and four diets with decreasing level of fishmeal (5% and 0%) with 10% Novacq™ and without (respective controls). The marine microbial biomass Novacq™ used in this study was produced in a 0.5Ha seawater-filled pond at the Bribie Island Research Centre in 2011 following a CSIRO patented production process (patent WO2009/132392). The manufacturing process and production of commercial quantities of Novacq™ is now licenced to Ridley Corp. in Australia. Novacq™ was incorporated by replacing an equal amount of wheat flour, adding 1% soybean meal to keep protein constant. Fishmeal was replaced by soybean meal and increasing levels of methionine, lysine and fish oil, with a concurrent decrease in soybean oil to keep total lipid constant (Table 2). The measured chemical composition confirmed all diets showed similar protein (34.5–36.1%) and energy content (18.8–20.6 KJ g⁻¹), including the benchmark commercial diet (Table 2).

Each of the ingredients were milled to < 750 µm before batching; ingredient chemical composition can be found in Table 1. A total of 10 kg of each of the experimental diets was mixed thoroughly (without

Table 1

Composition of the commercial diet and ingredients used in the experimental feeds, on dry matter basis.

Composition (%)	Commercial diet (CD)	Fishmeal	Soybean meal	Novacq™	Wheat flour	Wheat bran	Wheat gluten
Dry matter	83.3	90.0	90.0	93.0	87.6	89.2	90.2
Total protein	34.8	71.9	51.6	3.8	12.2	17.4	83.1
Total lipid	5.6	11.7	2.3	0.8	1.9	4.6	3.6
Gross energy (KJ g ⁻¹)	19.0	21.8	19.9	9.10	18.7	19.3	23.6
Ash	10.0	12.6	7.0	64.6	0.6	6.8	0.8
Carbohydrate	50.2	3.7	39.1	31.3	85.3	71.3	12.5

Table 2

Dietary formulations and composition of feeds on dry matter basis.

Ingredient (g kg ⁻¹)	10FM-0NQ	10FM-2.5NQ	10FM-5NQ	10FM-10NQ	5FM-0NQ	5FM-10NQ	0FM-0NQ	0FM-10NQ
Fishmeal	100	100	100	100	50	50	0	0
Soybean meal	310	320	320	320	380	390	450	460
Wheat bran	200	200	200	200	200	200	200	200
Novacq™	0	25	50	100	0	100	0	100
Wheat Flour	261	224	199	149	235	125	210	100
Wheat gluten	50	50	50	50	50	50	50	50
Fish oil	10	10	10	10	15	15	20	20
Soybean Oil	30	32	32	32	29	29	27	27
CaHPO ₄	20	20	20	20	20	20	20	20
L-Lysine	1	1	1	1	2	2	3	3
DL-methionine	5	5	5	5	6	6	7	7
Micronutrients*	13	13	13	13	13	13	13	13
Composition (%)								
Dry matter	93.7	94.1	94.4	95.1	94.7	94.2	94.4	94.8
Total protein	36.1	35.4	34.7	34.4	35.6	34.9	35.2	34.5
Total lipid	7.3	7.8	7.6	7.8	7.5	7.4	6.9	7.1
Gross energy (KJ g ⁻¹)	20.6	19.8	19.7	18.8	20.3	18.9	20.1	18.8
Ash	6.8	8.4	10.0	9.6	6.6	13.2	6.3	12.9
Carbohydrate	49.9	48.5	47.7	44.6	50.3	44.4	51.7	45.6

Fishmeal and Fish Oil: South American source, provided by Ridley, Narangba, QLD, Australia. Soybean meal: KEWPIE Stock feeds, Kingaroy, QLD, Australia; Wheat Flour and Wheat Gluten: Manildra, Auburn, NSW, Australia. Wheat Bran: Allora Grain and Milling, Allora, QLD, Australia.

* Micronutrients (g kg⁻¹ of diet): Choline (60% choline chloride), 5; Vitamin C (Stay-C), 2; fish vitamin and mineral premix, 6¹; ¹The vitamin and mineral premix content in the final feed (IU or g kg⁻¹ feed) was: Vitamin A, 15KIU; Vitamin D3, 1.5KIU; Vitamin E, 0.1 g; Vitamin K3, 0.01 g; Vitamin B1, 0.025 g; Vitamin B2, 0.03 g; Vitamin B3, 0.15 g; Vitamin B5, 0.05 g; Vitamin B6, 0.01 g; Vitamin B9, 5 mg; Vitamin B12, 0.03 mg; Biotin, 1 mg; vitamin C, 0.45 g; Choline chloride, 1 g; Inositol, 0.35 g; Ethoxyquin, 0.125 g; Copper, 0.015 g; Ferrous iron, 0.04 g; Magnesium, 0.1 g; Manganese, 0.09 g; Zinc, 0.15 g.

the oil component) using an upright planetary mixer (BakerMix, Artarmon, NSW, Australia). The diets were then extruded through a laboratory-scale 24 mm twin-screw extruder (MPF24:25, Baker Perkins, Peterborough, UK), with intermeshing, co-rotating screws. The screws were composed of a series of intermeshing feed screws (FS), forwarding paddles (FP) and lead screws (LS) arranged according to defined barrel diameters (D) such that overall configuration from the drive end was 16D FS, 2D FP, 1D FS, 2D FP, 1D LS, 1D FP, 2D LS to the die. A single 2.0 mm diameter cylindrical die tapered at a 67° angle with a land length of 3 mm was used. The barrel comprised of four temperatures zones that were set at 70, 80, 100 and 110 °C from the drive to die respectively, and the machine was run at c. 160 rpm. The mash was delivered into the barrel at a rate of c. 4.2–4.9 kg h⁻¹, and water was pumped peristaltically (Thermoline, Wetherill Park, NSW, Australia) into the barrel at between 2.1 and 3.0 L h⁻¹. Water addition was varied among diets to produce floating feeds of 2.5 ± 0.3 S.D. mm Ø. The pellets were cut to 3.9 ± 0.4 S.D. mm length at the die face with a 4 blade variable speed cutter. The pellets were then dried at 60 °C for 24 h, after which each diet was vacuum infused with the specific allocation of oil. The pellets were then allowed to cool before being bagged and labelled. All feed were kept frozen (–20 °C) except when feeding or weighing feed.

2.3. Feeding and growth measurements

Fish were fed by hand twice daily to apparent satiation. Each tank was fed over a minimum of three separate feeding events over 120 min in morning and 120 min in afternoon during which the fish were fed until uneaten pellets were observed for 10 min on the final occasion, indicating fish had reached satiation. Uneaten pellets were collected after each feeding, siphoned onto a 90 µm mesh screen daily, and dried in a pre-weighed aluminium foil until reaching a constant dry weight (4 h at 105 °C). Daily feed delivered was obtained from the differential weights before and after feeding for tank specific feed containers and expressed in dry matter. Feed intake was calculated daily as the difference between the dry matter delivered and dry matter collected. Feed intake (FI) was expressed in gram of feed dry matter per fish per day accounting for any daily mortality, as well as expressed as specific

feed intake (SFI), in percentage of body weight (i.e., SFI = FI/BW_e × 100). The average expected body weight (BW_e) of the fish each day was calculated based on fortnight fish weights and the measured linear growth rates achieved on all treatments.

$$\text{Feed intake (gDM/fish/day)} = \sum_{i=1}^{42} \frac{\text{Daily DM feed intake per tank}}{\text{Daily fish count}} / 42$$

Fish were weighed individually on days 0 and 42 and tanks were bulk weighed on days 14 and 28 to minimize fish handling stress. Fish were anaesthetized using clove oil before weighing and allowed to recover in their allocated tank following weighing. This procedure was carried out on each weight check day. Any observed mortality within the first week only was replaced by an equivalent fish of the same weight from the stock tank.

Fish growth was found to be linear over the experimental period for all treatments (on average, weight = 1.76 × Days + 7.49 g). Therefore fish growth was expressed as weight gain or daily weight gain in figures. Growth was also expressed in percentage difference to the control diet.

Feed conversion ratio (FCR) accounted for the mortalities and was also expressed correcting for feed dry matter, as follows:

$$\text{Food conversion ratio (FCR)} = \frac{\text{total feed intake}}{\text{total fish weight gain (inc. morts)}}$$

$$\text{FCR DM corrected} = \frac{\text{total DM intake}}{\text{total fish weight gain (inc. morts)}}$$

Nutrient retention efficiencies were calculated from feed intake and carcass composition for protein and energy as follows:

$$\text{Retention efficiency (RE;\%) of Nutrient} = 100 \times \left[\left(\frac{\text{Final fish nutrient content} - \text{average initial fish nutrient content}}{\text{average fish nutrient intake}} \right) \right]$$

2.4. Chemical composition analyses

Fish carcasses were frozen at –20 °C until analysis. Dry matter content was determined by gravimetric analysis following drying at

105 °C for 4 h. The samples were then ground, shipped to CSIRO analytical laboratories, and analysed for chemical composition. Ash content was determined based on mass change after combustion in a muffle furnace at 550 °C for 6 h. Total lipid content was determined gravimetrically following extraction in 2:1:0.4 chloroform:methanol:water, using a modification to the method proposed by Folch et al. (1957). Measurement of Total Nitrogen content was determined by the Dumas Method using a Flash 2100 Elemental Analyser (Thermo Fisher Scientific, Waltham, MA, USA) and used to calculate sample protein content based on $N \times 6.25$. Gross energy was determined by isoperibolic bomb calorimetry in a Parr 6200 oxygen bomb calorimeter with an 1108CL bomb for ingredients and diets, and an 1109A semi-micro bomb for faeces (Par Instrument Company, Moline, IL, USA). Carbohydrate was calculated by difference.

2.5. Statistical analyses

Differences in culture performance indices and carcass composition were tested by one-way ANOVAs followed by post-hoc comparisons using Tukey-Kramer tests. Repeated-measures ANOVAs were used to test differences in daily feed intake and specific feed intake between treatments for the entire duration of the experiment. Significant differences in daily weight gain with fishmeal and Novacq™ inclusions, and the interaction, were tested by a two-way ANOVA. Before all analyses, the ANOVA assumptions of normality of residuals and homogeneity of variances were tested using the Shapiro-Wilk and Levene tests, respectively. Survival data were arcsine square root transformed and data $\log_{10}$ or square root transformed when necessary to satisfy Levene's test for homogeneity of variance ($P > .05$). All statistical analyses were performed using NCSS 11.

3. Results

Juvenile tilapia grew well on the control experimental diet, with weight gain and survival not significantly different to a benchmark commercial diet (CD) (Table 3). Inclusion of Novacq™ increased weight gain significantly (Table 3 and Fig. 1). The average growth improvement against the control diet with no Novacq™ was 7.8%, 23.6% and 34.6% when included at 2.5%, 5% and 10%, respectively (Table 3). Significantly lower weight gain was achieved when fish were fed diets with 0% fishmeal compared to 10% fishmeal (Fig. 2). However, the addition of 10% Novacq™ had a positive effect on overall weight gain at all fishmeal inclusion levels (Fig. 2). The average growth against the control 10FM-ONQ decreased by 3% and 14.5% when fishmeal was reduced to 5% and 10%, respectively (Table 3). The average growth against the control 10FM-ONQ increased by 19.4% and 15.5% with the addition of 10% Novacq™ when fishmeal was reduced to 5% and 0%, respectively (Table 3).

Daily feed intake increased with Novacq™ inclusion and closely mirrored the growth trends (Fig. 1). Feed intake in absolute amount (FI)

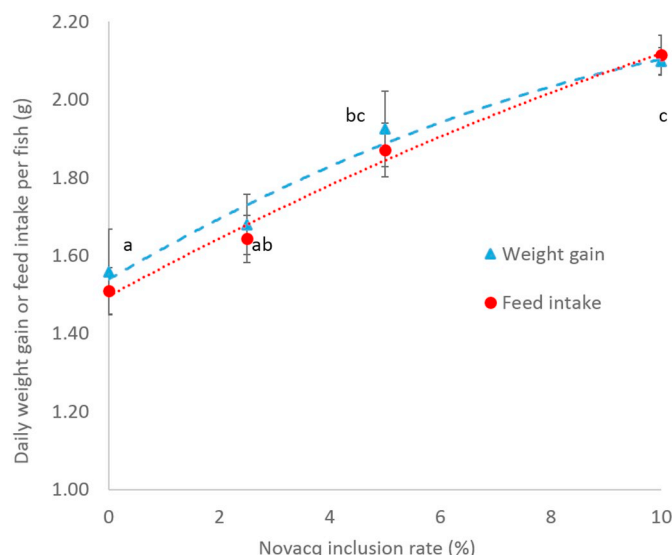


Fig. 1. Relationship between Novacq™ inclusion rate and the daily weight gain and daily feed intake of GIFT tilapia. Significant differences are marked by different letters [One-way ANOVA, daily weight gain, $P < .05$; daily feed intake, $P < .05$].

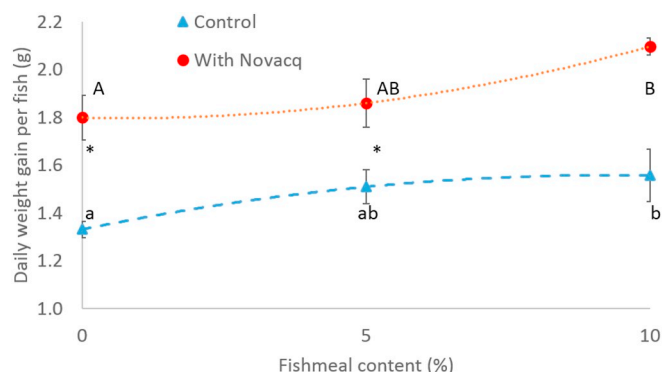


Fig. 2. Relationship between weight gain and fishmeal content of diets, with and without Novacq™ inclusion. Significant differences are marked by different letters [two-way ANOVA, Novacq™ effect, $P < .05$; fishmeal effect, $P < .05$; interaction, $P > .05$].

and specific to fish body weight (SFI) was significantly higher in the diets containing 10% Novacq™ than their respective control diets (i.e., 10FM-ONQ, 5FM-ONQ, 0FM-ONQ (Table 3). The trend for a higher specific intake with increasing inclusion of Novacq™ was also achieved consistently throughout the experiment (Fig. A.1).

Feed conversion ratios, as fed and expressed in feed DM, did not

Table 3

Summary of culture performance after 42 days of tilapia fed a benchmark commercial diet (CD), the experimental 10% FM diets with increasing inclusion of Novacq™, and experimental diets at 5% FM and 0% FM with and without 10% Novacq™ (mean $\pm$ S.E., $n = 6$).

	CD	10FM-ONQ	10FM-2.5NQ	10FM-5NQ	10FM-10NQ	5FM-ONQ	5FM-10NQ	0FM-ONQ	0FM-10NQ	Pooled SEM
Initial weight (g)	11.22	11.26	11.26	11.44	11.30	11.45	11.71	11.87	11.43	0.19
Weight gain (g)	69.17 ^{bcd}	65.43 ^{cd}	70.52 ^{bcd}	80.85 ^{ab}	88.09 ^a	63.43 ^{cd}	78.15 ^{ac}	55.94 ^d	75.59 ^{ac}	3.29
Growth diff. to 10FM-ONQ (%)	5.72 ^{bd}	0	7.77 ^{bc}	23.56 ^{ab}	34.63 ^a	-3.06 ^{cd}	19.44 ^{ab}	-14.51 ^d	15.52 ^{abc}	4.71
Survival (%)	80.00 ^b	83.33 ^{ab}	90.00 ^{ab}	93.33 ^{ab}	86.67 ^{ab}	91.67 ^{ab}	95.00 ^{ab}	86.67 ^{ab}	98.33 ^a	0.04
FI (g _{DM} fish ⁻¹ day ⁻¹)	1.74 ^{abc}	1.51 ^a	1.64 ^{ab}	1.87 ^{bcd}	2.11 ^d	1.53 ^a	1.96 ^{cd}	1.49 ^a	2.00 ^{cd}	0.06
SFI (%BW day ⁻¹)	5.06 ^{bcd}	4.50 ^a	4.46 ^a	4.76 ^{ab}	5.27 ^{cd}	4.50 ^a	5.10 ^{bcd}	4.83 ^{abc}	5.44 ^d	0.10
FCR	1.31 ^a	1.05 ^b	1.05 ^b	1.04 ^b	1.08 ^{bc}	1.08 ^{bc}	1.13 ^{cd}	1.15 ^{cd}	1.18 ^d	0.03
FCR _{DM}	1.09 ^{ab}	0.99 ^b	0.99 ^b	0.98 ^b	1.02 ^{ab}	1.02 ^{ab}	1.06 ^{ab}	1.09 ^{ab}	1.12 ^a	0.03
RE _{protein}	43.26	45.44	45.23	45.62	44.17	45.02	40.30	38.16	36.97	3.17
RE _{energy}	26.36	26.97	25.44	30.55	28.14	25.59	23.50	23.53	22.79	2.45

* Significant differences are marked by different letters [One-way ANOVAs, $P < .05$; Repeated-measures ANOVA for FI and SFI, $P < .05$].

Table 4

Mean carcass composition of tilapia on a wet weight basis after 42 days fed a benchmark commercial diet (CD), the experimental 10% FM diets with increasing inclusion of Novacq™, and experimental diets at 5% FM and 0% FM with and without 10% Novacq™.

	CD	10FM-0NQ	10FM-2.5NQ	10FM-5NQ	10FM-10NQ	5FM-0NQ	5FM-10NQ	0FM-0NQ	0FM-10NQ	Pooled SEM
Dry matter (%)	26.53	26.58	25.14	26.84	26.09	27.18	24.28	25.67	23.69	1.33
Crude protein (%)	15.76	15.40	15.20	15.00	15.07	15.54	14.42	15.76	13.91	0.79
Gross energy (KJ/g)	6.48	6.47	5.99	6.67	6.22	6.29	5.62	6.15	5.70	0.33
Total lipid (%)	7.28	7.27	6.32	7.83	6.90	6.69	6.43	7.43	6.51	0.42
Ash (%)	3.42	3.58	3.71	3.50	3.63	3.91	3.37	3.17	3.23	0.28

* No significant difference was found [One-way ANOVAs, $P > .05$].

vary significantly with Novacq™ inclusion against the respective experimental controls. FCRs were significantly better on all experimental feeds than the benchmark CD, but FCR_{DM} reduced considerably for CD to become not significantly different (Table 3), as moisture content was higher for CD (Table 1). FCR and FCR_{DM} tended to increase with a reduction in fishmeal but did not vary significantly with fishmeal and/or Novacq™ inclusion (Table 3). Retention efficiency of protein and energy did not differ significantly between diets (Table 3). Carcass composition analyses indicated no significant differences between fish (Table 4).

4. Discussion

This study was carried out to answer two main hypotheses, 1) Is there a growth benefit to including Novacq™ at various inclusion rates in practical extruded tilapia diets, and 2) can Novacq™ improve the growth performance of tilapia in diets with low to no fishmeal. Our results have clearly verified both hypotheses with the inclusion rate of Novacq™ at 10% significantly increasing growth performance of tilapia by up to 35% in diets with 10%, 5% and 0% fishmeal. These growth benefits are comparable to previously achieved in the prawn *P. monodon* (Glencross et al., 2014, 2015; Arnold et al., 2016). Furthermore, our diets were extruded indicating the benefit of Novacq™ is not lost when common extrusion heat treatment is applied. All extruded diets containing Novacq™ outperformed a benchmark commercial diet containing similar amount of protein and energy, so growth results achieved in our study are commercially relevant. Weight gain increased in parallel with increasing inclusion rates of Novacq™, with weight gains of 7.8%, 23.6% and 34.5% in relation to an isoenergetic and isonitrogenous control diet, at 2.5%, 5% and 10% Novacq™ inclusion, respectively. While the weight gain response slowed towards the highest inclusion, we did not achieve a response plateau in this study indicating greater weight gains may be achievable at higher inclusion rates in tilapia.

Total replacement of 10% fishmeal with soybean meal and increased inclusion of methionine, lysine and fish oil to compensate for nutrient deficiencies, resulted in a significant decrease of around 14.5% in daily weight gain. An intermediate level of fishmeal (5%) did not significantly affect weight gain. Our observed reduction in growth on a 0% fishmeal diet could not be linked to a significant impact of removing fishmeal on feed intake ($4.8\% \text{ BW day}^{-1}$), FCR (1.06), protein and energy retention. These results are in agreement with several previous studies showing that tilapia growth performance reduces with the replacement of fishmeal for individual plant proteins, possibly due to the presence of anti-nutritional factors, particularly from soybean meal (Furuya et al., 2004; Borgeson et al., 2006; Koch et al., 2016). Here we demonstrate that Novacq™ can increase the weight gain of tilapia on 0% fishmeal diets by as much as 35%, resulting in a net improvement in culture performance of 15.5% against the 10% fishmeal control. As such, the use of Novacq™ as a feed additive can more than compensate for the negative impact of eliminating fishmeal in diets for tilapia.

It is important to assess ingredient palatability by measuring feed intake, and for an animal to demonstrate an accurate feed intake response it must be given the opportunity to refuse feed (Glencross et al.,

2007). In this study, careful hand feeding allowing sufficient time and effort to feed fish to satiety was performed in order to collect uneaten food from all tanks twice daily after the feeding events. Feed intake on the experimental diets mirrored the growth trends achieved. However at this point it is difficult to determine if the fish were growing faster because they were eating more and/or were eating more because they were growing faster. We attempted to decouple growth and feed intake by expressing feed dry matter intake as a percent of daily expected body weight based on measured linear growth rates, to account for fish weight variation. Fish fed the commercial benchmark diet showed a relatively high dry matter intake in relation to their poorer growth rates when compared to our experimental diets. This resulted in significantly poorer FCR, possibly due to the use of less digestible ingredients, although FCR expressed in dry matter improved due to the higher moisture content of the commercial diet compared to our experimental diets. Fish fed Novacq™ diets ate significantly more food over the course of the experiment than their respective control diets fed fish, both in absolute terms but also in percent of their body weight, and the response was in a direct relationship with inclusion rates. A 10% inclusion of Novacq™ resulted in approximately 33% increase in feed intake at all levels of fishmeal inclusion, indicating a clear effect of this feed additive on tilapia diet palatability regardless of fishmeal content. The effect of Novacq™ as a feeding stimulant was rapid and observed within the first few days of feeding, with fish displaying a voracious feeding behaviour and greater appetite compared to counterparts fed control diets (Pers. Obs.). Similar FCRs and REs measured in this study point towards Novacq™ acting principally as a feeding stimulant in tilapia. Further research to confirm the effect is warranted, through the use of palatability inhibitors and limiting-constraint trials (Glencross et al., 2007). Additionally, characterising feed digestibility and the type of chemical stimuli originating from marine microbial biomass is of interest, as it is possible that other positive effects of Novacq™ were not identified in this study. Considering feed intake could be stimulated by Novacq™ despite its low nitrogen content (i.e., $N = 0.6\%$), the results were unexpected and possibly indicate stimulation of different gustatory or metabolic pathways to those linked to typical feeding attractants and stimulants (de la Higuera, 2007; Lamb, 2007).

The use of Novacq™ to improve the palatability and growth rates of tilapia juveniles on feeds with low to zero fishmeal, and soybean meal as the major protein source, is a positive finding. It alleviates the need for some wild-caught fishery products and greatly improve feed sustainability. Further research is warranted on the potential for Novacq™ to assist in incorporating more locally available feed grade raw ingredients and its cost-effectiveness in commercial rearing conditions for tilapia and other fish.

Acknowledgment

This study was funded by the Department of Foreign Affairs and Trade, Australian Government, under the InnovationXchange Blue Economy Challenge program. We thank Trini Pratiwi for maintenance of the experiment, Barney Hines for sample importation and processing, Sue Cheers and Nicholas Bourne for sample analyses.

Appendix A

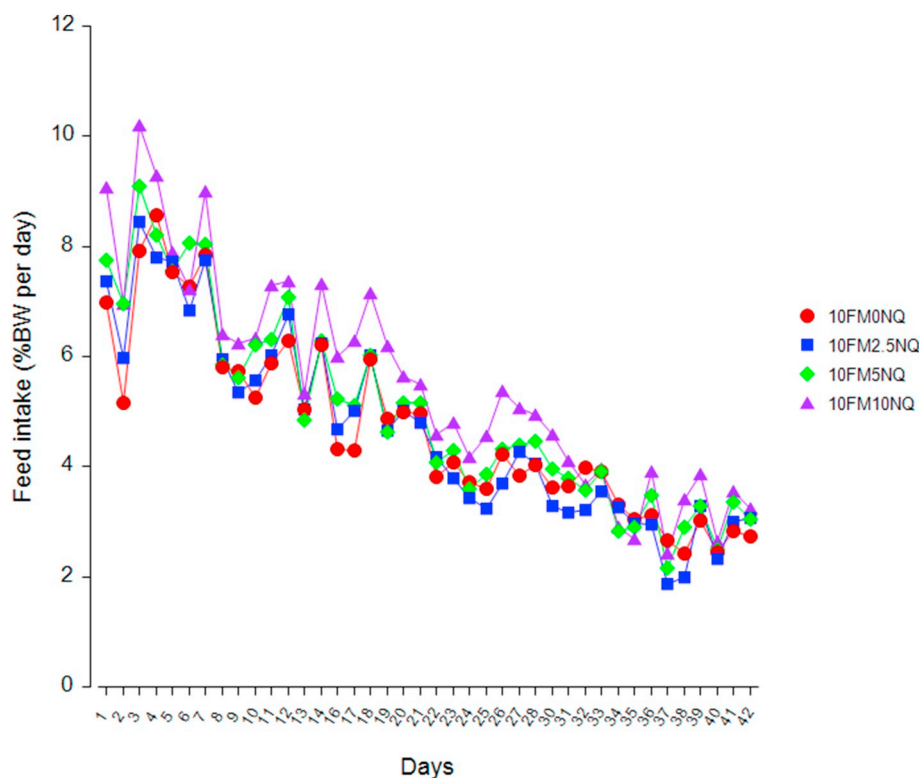


Fig. A.1. Mean daily dry matter feed intake for different inclusion rates of Novacq™, expressed in % of body weight, over the course of the experiment.

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